8.6.2 Ergodicity

Following the discussion in [220], let us show ergodicity for the stochastic isokinetic harmonic oscillator:

$$\begin{aligned} \mathrm{d}q &= p\mathrm{d}t \\ \mathrm{d}p &= [-q - \lambda p]\mathrm{d}t \\ \mathrm{d}\xi_1 &= [-\lambda\xi_1 - \xi_2\xi_1]\mathrm{d}t \\ \mathrm{d}\xi_2 &= [\xi_1^2 - \mathrm{k_B T} - \gamma\xi_2]\mathrm{d}t + \sigma\mathrm{dW} \end{aligned} \tag{8.45}$$

with $\lambda = \frac{-qp-\xi_2\xi_1^2/2}{p^2+\xi_1^2/2}$ and $\sigma = \sqrt{2\gamma \mathrm{k_B T}}$. We have taken the coupling coefficients μ_1, μ_2 both one for simplicity as their particular values (as long as nonzero) have no bearing on the ergodic property.

The constraint, $p^2 + \xi_1^2/2 = \text{const}$, defines a three-dimensional manifold $\mathcal{M}$, a generalized cylinder. $\xi_1 = 0$ is an invariant manifold of the isokinetic equations, with solutions therefore confined for all time to one or the other half-space ($\xi_1 < 0$ or $\xi_1 > 0$). It is enough to sample one or the other of the two half spaces. To show that the solutions do so, we need to demonstrate that an appropriate Hörmander condition holds. Define $\boldsymbol{f}$ and $\boldsymbol{g}$ as the vector fields

$$\boldsymbol{f} = p\partial_q - (q + \lambda p)\partial_p - (\lambda\xi_1 + \xi_2\xi_1)\partial_{\xi_1} + \left[(\xi_1^2 - \mathrm{k_B T}) - \gamma\xi_2\right]\partial_{\xi_2}, \tag{8.46}$$

and

$$\boldsymbol{g} = \sigma\partial_{\xi_2}. \tag{8.47}$$

We will show that the dimension of the ideal spanned by the iterated commutators of the vector fields $\boldsymbol{f}$ and $\boldsymbol{g}$ is three, i.e. that it spans the tangent space to $\mathcal{M}$ at any point. For this purpose, we use the collection of vectors

$$\boldsymbol{f}, \boldsymbol{g}, [\boldsymbol{f}, \boldsymbol{g}], [\boldsymbol{f}, [\boldsymbol{f}, \boldsymbol{g}]], [\boldsymbol{f}, [\boldsymbol{f}, [\boldsymbol{f}, \boldsymbol{g}]]].$$

We will see that these provide a basis at every point excluding the one-dimensional set defined by the union $\mathcal{M}_0$ of the two lines ℓ_+, ℓ_- defined by

$$q = 0, \qquad p = 0, \qquad \xi_1 = \pm\sqrt{2\mu_1^{-1}\mathrm{k_B T}},$$

and arbitrary ξ_2. [Note: the choice $\xi_1 = \pm\sqrt{2\mu_1^{-1}\mathrm{k_B T}}$ is made so that the isokinetic constraint is satisfied. In [220] this value was incorrectly defined.] The vectors $\boldsymbol{f}$ and $\boldsymbol{g}$ are linearly independent except on $\mathcal{M}_0$.